

Linear Regression (House Price Prediction)

Objective:

The goal of this task is to build a Linear Regression model to predict house prices using the California Housing dataset. The dataset contains multiple numerical features such as income, house age, number of rooms, population, and location-based information.

Dataset Used

California Housing Dataset (Scikit-learn)

- Features: 8 numerical columns
 - Target: MedHouseValue (continuous numeric variable representing median house value)
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Steps Performed

1. Loaded the dataset using `fetch_california_housing()` and converted it into a Pandas DataFrame.
 2. Performed dataset inspection using `.head()`, `.info()`, and `.describe()` to understand ranges and missing values.
 3. Split the data into:
 - **X (features)**
 - **y (target: MedHouseValue)**
 4. Performed train-test split using:
 - 80% training data
 - 20% testing data
 - `random_state=42`
 5. Trained a **Linear Regression** model using `LinearRegression()` from scikit-learn.
 6. Predicted house values on the test dataset.
 7. Evaluated the model using **MAE** and **RMSE**.
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Evaluation Metrics

1. Mean Absolute Error (MAE)

MAE measures the average absolute difference between actual and predicted values.

Interpretation:

Lower MAE means predictions are closer to actual values.

2. Root Mean Squared Error (RMSE)

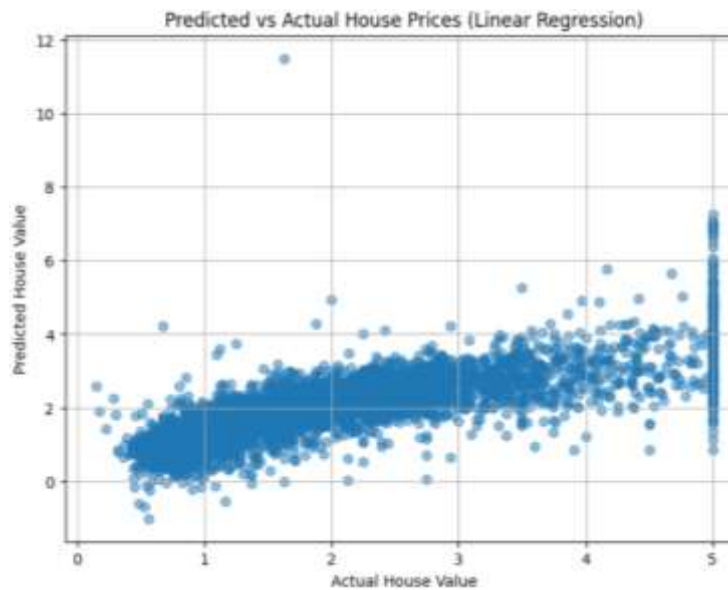
RMSE is the square root of the average squared errors.

Interpretation:

RMSE penalizes larger errors more heavily than MAE, making it important when large mistakes are costly.

Results

- **MAE**
- **RMSE**



Conclusion:

The Linear Regression model provides a baseline regression performance. The predicted vs actual scatter plot shows how closely predictions align with real values. While the model performs reasonably, errors exist due to the limitation of Linear Regression in capturing non-linear relationships.

Key Learning

- MAE is easier to interpret (average absolute error).
- RMSE highlights large prediction mistakes more strongly.
- Linear Regression coefficients help identify which features impact house prices the most.